

# CS1302 Introduction to Computer Programming

## Course overview

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# About the course

This course aims to introduce key concepts, techniques, and good practices of programming using a high-level programming language Python.

## Course Intended Learning Outcomes (CILOs):

1. Explain the structure of a computer program.
2. Analyze, test and debug computer programs.
3. Apply proper programming techniques to solve a task.
4. Construct well-structured programs.

# About the course

- **Lecturers**

Prof. Chung Chan, [chung.chan@cityu.edu.hk](mailto:chung.chan@cityu.edu.hk) (Course leader, Course management)

Prof. Weitao Xu [weitaoxu@cityu.edu.hk](mailto:weitaoxu@cityu.edu.hk) (Teaching)

Prof. Minjing Dong, [minjdong@cityu.edu.hk](mailto:minjdong@cityu.edu.hk) (Teaching)

- **TAs**

- We have about 20 TAs to offer assistance on teaching and QA sessions.

- **Teaching mode**

- Lecture : On-campus face-to-face

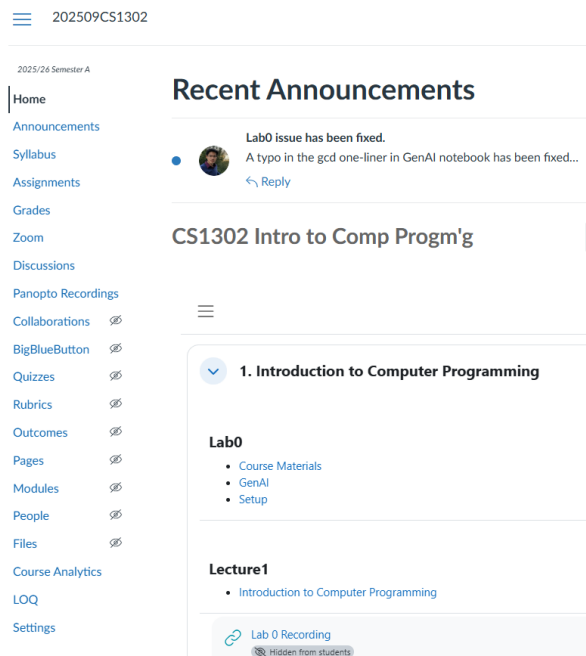
- Labs: On-campus face-to-face

Lectures/Labs will be recorded on Zoom, but the recording is used for review purpose. The recording will be unavailable one week after it is published. You're encouraged to attend the class physically.

Attendance will not be marked, but individual lectures may record attendance for progress check.

# Course materials

- Course contents are delivered in the Jupyterhub platform
- In Lab0, you will learn how to set up your account



- Textbook

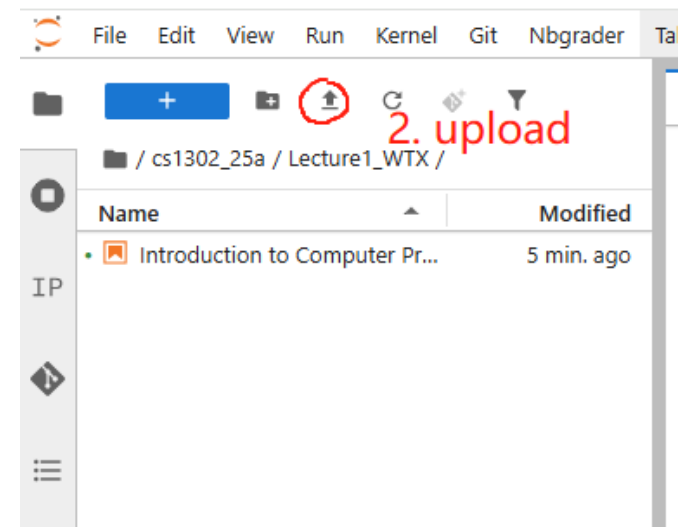
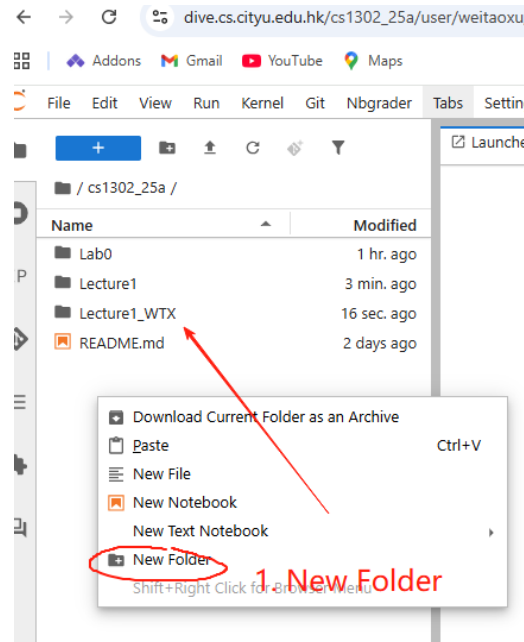
Halterman, R. L. "Fundamentals of Python programming." (2017)

- [E-Book available from openlibrary.org \(Links to an external site.\)](https://openlibrary.org)
- [Source code available from GitHub repository](#)

# Course materials

- But, I prefer to use my materials, which add some content based on the official ones.
- Where to find?
  - Go to my website: <https://www.weitaoxu.com/resources/>

- How to use?



# Course materials

- E-quiz: just for practice, no marks

CS1302 Intro to Comp Progm'g

Assign To Edit

The sidebar shows three sections: 'Reading' with a link to '[Halterman17] 1.1-1.3', 'Supplements' with a link to 'E-Quiz 1', and a notification 'How should we credit Alan Turing?'. Each section has a 'Completion' dropdown menu. A red arrow points from the 'E-Quiz 1' link to the text below.

## Recent Announcements

CS1302 Intro to Comp Progm'g

The sidebar is shown with a right-click context menu open over the 'E-Quiz 1' link. The menu options are: 'Open link in new tab', 'Open link in new window', 'Open link in incognito window', 'Open link as Weita (gapps.cityu.edu.hk)', 'Save link as...', 'Copy link address', and 'Inspect'. A red arrow points from the text below to the 'Open link in new tab' option.

The window is too small?  
Try to right-click, and “Open link  
in new tab”

# Assessment

- **Coursework (50%) :**

1. Lab assignments: 30%

2. Midterm Quiz: 20%

- Date: 18-Oct-2023 (Saturday) 18:30-19:30, mark it on your calander
- Computer-based: you will use Safe Exam Browser on the provided computers to access the exam (without connection to Internet during exam except Canvas).

- **Final Exam (50%)**

The final exam will be on computer, and the duration is 2 hours.

Passing criteria:

At least 30% of the maximum mark for the examination must be obtained.

- The distribution of scores is different every year, so it's hard to tell the boundaries of A+/A/A-/B+/B/B-.
- The final grade is determined by the relative position of your score (top 3%, 20%, 50%) but not the absolute value of your score (90, 85, 70..)

# Academic Honesty

is regarded as a serious academic offence in the University.

Do assignments and exercises **on your own**

“**On your own**” means

- ✓ discuss the problems with any other people.
- ✓ study materials available on the internet.
- ✓ refer to books.

**But the details and write-up must be entirely your work.**

The **principle** is: **Students should gain through practicing and developing skills in doing your work.**



# Academic Honesty

is regarded as a serious academic offence in the University.

Do assignments and exercises **on your own**

**Deserved mark?**

**Unfair situation** ✕

**You should not create any chance for other students to copy your work.**

**For any plagiarism case,**

- The student who plagiarizes will be punished.
- Any student who allows his/her work to be copied will **also** be punished.

# Do students in the following cases demonstrate academic honesty?

## Case 1

- Chi Wai did some research on the Internet and found some useful sources for his assignment. He used the ideas from these sources but forgot to cite the sources in the text of his assignment and provide a reference list.

## Case 2

- Mia searched the Internet and found a useful diagram. She modified the diagram and included it in her assignment, but did not cite and reference the source.

## Case 3

- Nick handed in the same assignment for more than one subject/course.

## Case 4

- Xiaobo copied his assignment from a classmate's work, put his own name on it and submitted it to his teacher.

## Case 5

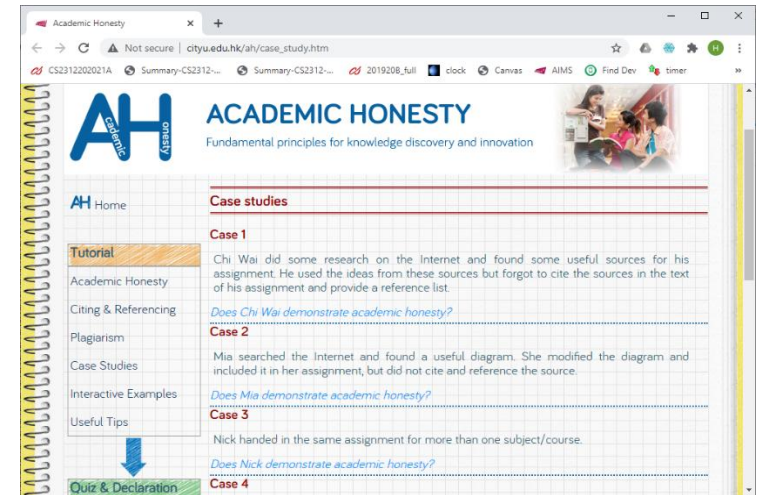
- Kit gave Suki his assignment and Suki copied it, put her own name on it, and submitted it to her teacher.

## Case 6

- Chloe and Eason worked on an individual assignment together. They each did a different part of the assignment. Then, they shared their work, put their own name on the assignment and individually submitted the work to their teacher.

## Case 7

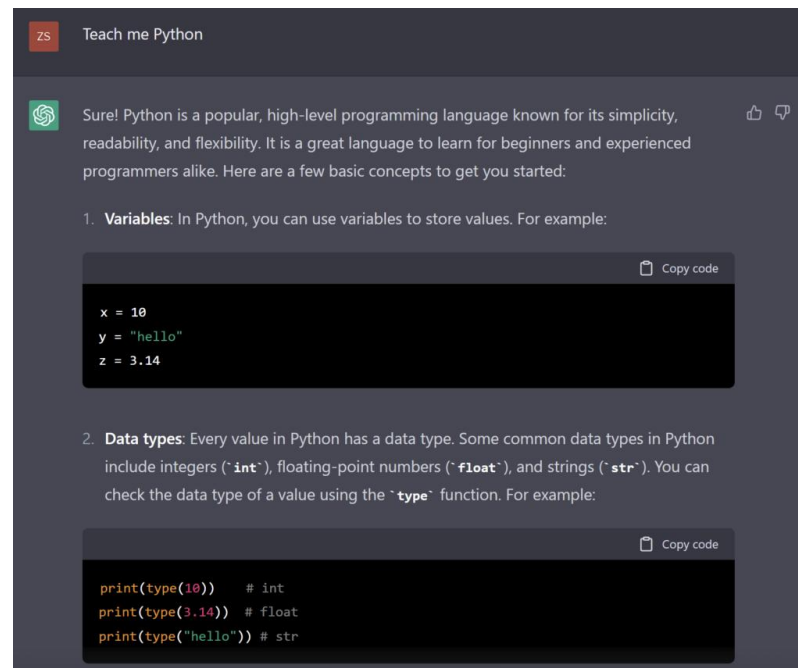
- Ming and Doris discussed some ideas about how to do an individual assignment. Then, they completed the assignment on their own, without looking at each other's work. They put only their own name on their assignment and individually submitted their work to their teacher.



<https://www.cityu.edu.hk/ah/Tutorial/casestudy>

# Can I use ChatGPT

- You're free to use any tools that can help your study, but you won't have access to Internet during the midterm/final exams



# About the course – Key to Success

## Practice with understanding

**The solution is not everything.**

You need to learn **the approach** to build the solution step by step.

## Learn from errors (**debugging strategies**)

Mistakes are often **good teachers** 💰

## **Digest** well to learn the 100%

Memorizing and dictation,  
if without understanding,  
are worthless,  
especially in a programming course!

# Python is a tool to solve questions

In some courses, what you learn  
in the class

$$\begin{aligned}x^2 + 2 &= 18 \\ x^2 &= 18 - 2 \\ x &= 4\end{aligned}$$

similar



What you see in the exam

$$\begin{aligned}x^2 + 9 &= 100 \\ x^2 &= 100 - 9 \\ x &= 8\end{aligned}$$

In this course, what you learn in  
the class

**Python for loop**  
A for loop is **used for iterating over a sequence and iterables** (like range, list, a tuple, a dictionary, a set, or a string).

```
for i in range(5):  
    statement 1  
    statement 2  
    ...  
    statement n  
else:  
    statement(s)
```

**Indentation**  
Loop body is must be properly indented

**Definite iterations.**  
(Total 5 iterations)

**Body of for loop**  
Execute till the last item of a sequence

**Else Block (optional)**  
Execute only when for loop executes normally

PYnative.com

unseen



**Example 1: Program to print half pyramid using \***

```
*  
* *  
* * *  
* * * *  
* * * * *
```

**Source Code**

```
rows = int(input("Enter number of rows: "))  
  
for i in range(rows):  
    for j in range(i+1):  
        print("* ", end="")  
    print()
```

# After this course

Become a Programmer



Cool and fun

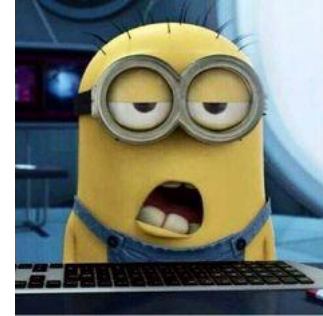


But being a programmer (程序猿)...



# What to expect

- Programming is fun
- *Learning programming may not!*
- You'll see lots of syntax (rules for programming languages), yes they may look boring...
- You'll see lots of errors, that'll take you lots of time to fix...



```
turnonlight.py | pi@raspberrypi: ~/notebook X | turnonlight.py |  
+ + + + + Code  
[1]: import RPi.GPIO as GPIO # Import Raspberry Pi GPIO library  
from time import sleep # Import the sleep function from the time module  
GPIO.setwarnings(False) # Ignore warning for now  
GPIO.setmode(GPIO.BOARD) # Use physical pin numbering  
GPIO.setup(4, GPIO.OUT, initial=GPIO.LOW) # Set pin 4 to be an output pin and set initial value to low (off)  
  
while True: # Run forever  
    GPIO.output(4, GPIO.HIGH) # Turn on  
    sleep(1) # Sleep for 1 second  
    GPIO.output(4, GPIO.LOW) # Turn off  
    sleep(1) # Sleep for 1 second  
  
.....  
ModuleNotFoundError: Traceback (most recent call last)  
<ipython-input-1-b5e23991a7c5> in <module>  
----> 1 import RPi.GPIO as GPIO # Import Raspberry Pi GPIO library  
      2  
      3 from time import sleep # Import the sleep function from the time module  
      4  
      5 GPIO.setwarnings(False) # Ignore warning for now  
ModuleNotFoundError: No module named 'RPi'
```

# What to expect

- To have fun, usually we need to go through lots of detail stuff first -> learning curve

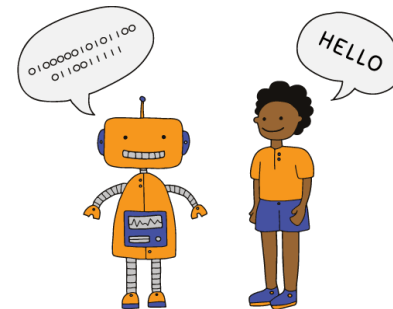


## Positive thinking

- You learn a language .. What are the stories behind the details? You *communicate* with the language.
- You create solutions and solve mistakes
  - You are winning the games!

# One more tip

- If the computer says you're wrong, you are wrong!  
Don't doubt the computers, ever. They are always right  
(unlike CNN or Trump...)



- Focus on finding out what make the computer think you're wrong
  - Debug
  - [Why are they called bugs?](#)



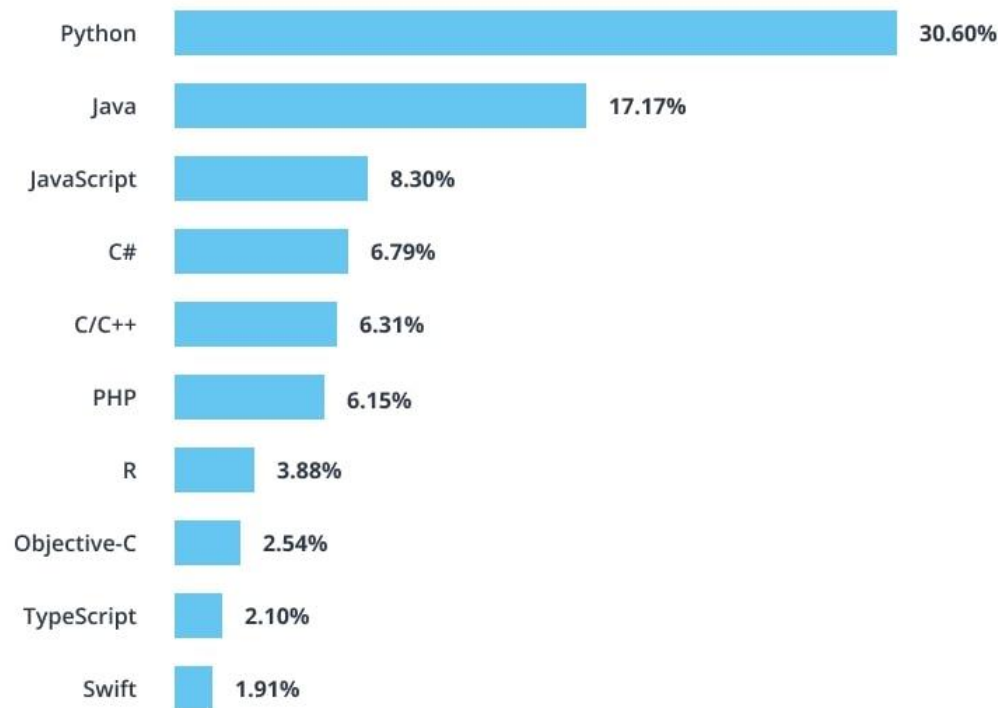
# There are many Programming Languages in the world

Ada Assembly Basic C C++ C# Cobol  
Cobra CODE ColdFusion Delphi Eiffel  
Fortran FoxPro GPSS J# J++ Java  
JavaScript LISP Logo LUA MEL Modula-2  
Miranda Objective-C Perl PHP Prolog  
Python Go Visual Basic Swift

# Programming Language

Python:

The most popular programming language



# Programming Language

...

What programming language is mostly used in the area of AI? Why?



GPT-3.5-Turbo

Poe

...

Python is the programming language that is predominantly used in the field of AI. There are several reasons for this:

1. **Simplicity and Readability:** Python is known for its simplicity and readability. Its syntax is clear and concise, making it easier for developers to write and maintain code.
2. **Rich Ecosystem:** Python has a vast ecosystem of libraries and frameworks that are specifically designed for machine learning and AI, such as TensorFlow, PyTorch, and scikit-learn. These libraries provide tools for tasks like data manipulation, mathematical operations, and model building.
3. **Community Support:** Python has a large and active community of developers. This means that there are plenty of resources, tutorials, and forums available for AI practitioners to seek help and share knowledge.
4. **Flexibility:** Python is a versatile language that can be used for a wide range of applications beyond AI. This flexibility makes it a popular choice for developers who work on diverse projects.
5. **Performance:** While Python may not be the fastest language for certain tasks, its performance can be optimized using libraries like NumPy and by leveraging its integration with high-performance languages like C and C++.

Overall, Python's ease of use, extensive libraries, community support, and flexibility make it the preferred choice for AI and machine learning applications.

# Programming Language

Almost all the open-sourced ML/DL projects are built upon **Python**.

For example:

## 1. Segment Anything

- <https://segment-anything.com/demo>

## 2. Stable Diffusion

- <https://huggingface.co/spaces/stabilityai/stable-diffusion-3-medium>

## 3. Music Generation

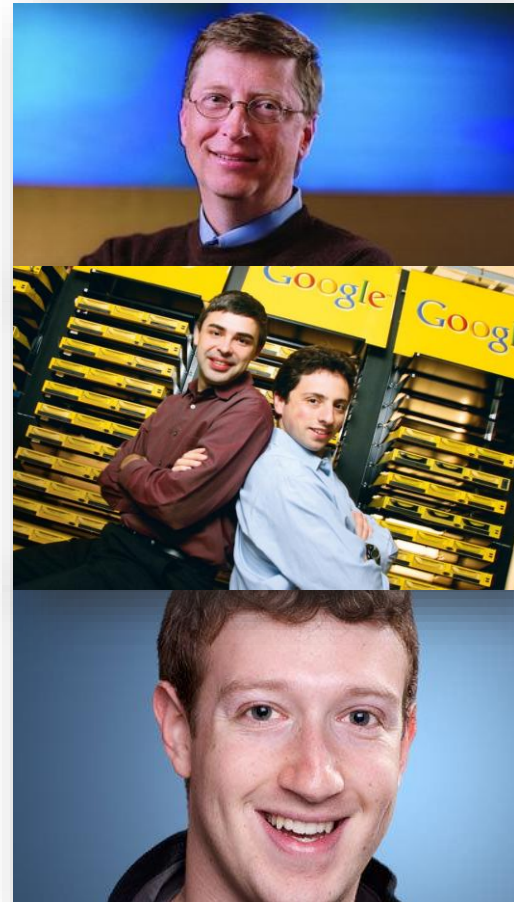
- <https://musicgeneratorai.com/>

## 4. Pose Estimation

- <https://saiwa.ai/app/detection/pose-estimation-1/>

## Average Python Developer Salaries

|             |           |
|-------------|-----------|
| Switzerland | \$133,000 |
| The USA     | \$127,000 |
| Denmark     | \$81,800  |
| Israel      | \$71,000  |
| Norway      | \$62,000  |
| Czech       | \$34,122  |
| Poland      | \$21,552  |
| Hungary     | \$16,771  |
| Ukraine     | \$15,906  |



## Look for more practice?

- They are plenty of online programming resources, but you need to choose the one suitable for this course.
- <https://www.geeksforgeeks.org/python-exercises-practice-questions-and-solutions/?ref=shm>
- <https://www.placementpreparation.io/mcq/python/>
- <https://pynative.com/python-exercises-with-solutions/>
- <https://www.w3resource.com/python-exercises/>
- <https://pynative.com/python-basic-exercise-for-beginners/>
- Or ask ChatGPT to create some programming questions for you.